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FINAL EXAMINATION
SEMESTER 2 SESSION 2017/2018

CODE: MES3103 **COURSE: SOFTWARE VERIFICATION AND VALIDATION**

DATE: 04 JUN 2018 **DURATION: 2 HOURS**

INSTRUCTIONS

1. This exam paper consists of **ONE (1)** part only, namely **Part A**.
2. Answer **ALL** questions in the answer booklet provided.
3. This question paper will be collected at the end of the examination.

This question paper consists of 3 pages including cover

PROGRAMME: _____

YEAR: _____ LECTURE GROUP: A / B

REGISTRATION NO.: _____ | | | | | | |

IDENTITY CARD NO.: | | | | | | | | | | | | |

LECTURER: DR. AZNIAH BINTI ISMAIL

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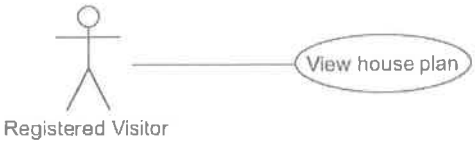
PART A: (80 marks)

Instruction: Answer **all** questions in the answer booklet provided.

1. Explain the differences between *verification* and *validation* (in terms of WHAT, WHEN, WHERE, WHO and HOW).

[20 marks]

2. Given the following use case diagram and its use case description, (Figure 1) create an appropriate test case that can be used during the validation process. Follow the format given in *Table 1*.



Use case ID	TBR03
Use case Name	View House Plan
Author	Siti Fatimah Sezali
Date	04/01/2017
Actor	Registered Visitor
Description	This function is only visible to the Registered Visitors. Once logged in, a Registered Visitor shall be able to view all house plans that are provided in the TBR system by clicking the View House Plan button in the Registered Visitor's main page.
Precondition	TBR system is ACCESSIBLE at: Https://www.tamanbahterareservation.com. Registered Visitor has successfully logged in into the system.
Basic steps	1. The main page for Registered Visitor is displayed. 2. Registered Visitor clicks View House Plan button in the menu. 3. System displays the house plan page. 4. System retrieves all house plan images and details from database. 5. System displays all retrieved house plan images and details in the house plan page.
Alternate steps	N/A
Exceptions	E4.1 Cannot access database E4.1.1 System display a pop-up notification of error. E4.1.2 Registered Visitor may try again.
Business validation/Rules	House plans can be accessed by Registered Visitor only. Registered Visitor must have an existing account with TBR.
Post conditions	Registered Visitor can view all the house plan images and read their details by scrolling the page. Registered Visitor may log out or go back to main menu at any time.

Figure 1 : View House Plan Use Case

Table 1 Test case form

Test Case ID:						
Test Case Name:						
Step	Procedure	Expected Result	Actual Result	Pass/Fail	Tester Name	Date

[20 marks]

[See Next Page

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3. Explain the differences between *defect*, *error*, *bug*, *failure* and *fault*.

[20 marks]

4. Software may fail without proper verification and validation. However, it can be a great problem to perform proper verification and validation. Briefly explain **FIVE (5)** reasons on how such problem could happen.

[20 marks]

END OF QUESTIONS

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